

LONG ISLAND BUOY 44025, USA

WMO: 996420

Lat: 40.251N

Lon: 73.164W

Elev: 0

StdP: 14.70

Time Zone: -5.00 (W05)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.1	22.5	4.4	6.9	21.7	8.0	8.3	23.8	38.9	27.3	36.6	31.0	18.6	290	0.526

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.7	78.5	74.7	77.6	74.1	76.5	73.1	76.6	77.7	75.3	76.8	74.3	75.8	10.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.1	136.1	77.4	74.8	130.3	76.4	73.6	125.2	75.4	40.0	78.0	38.8	77.0	37.7	75.9	79.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
34.7	31.2	26.3	15.5	79.6	5.1	1.8	11.8	80.9	8.9	81.9	6.0	82.9	2.4	84.2		
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.9	36.7	36.9	40.4	46.8	54.7	64.9	72.5	72.4	68.0	58.9	50.6	43.2
	DBStd	13.86	8.31	6.19	4.86	4.09	4.37	4.76	3.08	3.10	3.76	4.86	6.16	7.28
	HDD50	1490	413	366	297	108	6	0	0	0	0	4	66	230
	HDD65	4669	877	786	762	545	319	60	0	0	16	194	433	677
	CDD50	2924	1	0	0	14	153	447	696	694	539	279	83	18
	CDD65	626	0	0	0	0	1	56	231	229	104	5	0	0
	CDH74	1053	0	0	0	0	0	13	507	482	51	0	0	0
	CDH80	2	0	0	0	0	0	0	2	0	0	0	0	0
Wind	WSAvg	12.3	15.3	14.7	13.4	11.2	9.9	9.8	9.3	9.0	10.7	13.4	14.6	15.9
Precipitation	PrecAvg	45.8	3.5	3.0	4.2	3.9	3.8	3.9	3.7	4.5	3.8	3.9	3.4	4.3
	PrecMax	57.4	8.1	5.8	8.8	8.7	8.0	9.0	6.5	13.4	9.0	15.2	5.8	7.4
	PrecMin	36.5	1.6	1.0	1.1	1.6	1.3	0.9	1.0	0.7	1.4	0.3	1.3	1.3
	PrecStd	5.7	1.3	1.3	1.8	1.7	1.4	2.0	1.5	2.5	1.9	2.9	1.3	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.2	50.5	52.2	58.5	68.4	75.8	80.2	79.5	77.2	70.5	63.1	59.2
		MCWB	52.8	48.9	50.4	54.4	65.3	71.4	75.8	75.2	74.8	68.1	61.0	57.4
	2%	DB	51.8	48.2	50.1	55.3	65.1	73.9	79.0	78.4	75.5	68.6	61.5	57.1
		MCWB	50.0	46.2	48.0	51.9	62.9	70.8	75.1	74.6	72.6	66.1	59.0	55.1
	5%	DB	49.8	46.3	48.1	53.2	62.5	72.7	77.9	77.7	74.1	66.9	60.1	55.5
		MCWB	47.7	44.1	46.0	49.9	60.8	69.8	74.3	74.2	70.9	63.5	57.7	53.4
	10%	DB	47.8	44.6	46.4	51.6	60.3	71.1	76.8	76.7	73.0	65.5	59.0	53.6
		MCWB	45.0	41.9	44.2	48.5	58.5	68.3	73.3	73.2	69.5	61.8	56.5	51.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.7	50.1	51.4	56.2	66.7	73.7	77.8	77.8	76.3	69.5	61.7	58.4
		MCDB	54.2	50.7	51.6	57.8	67.3	74.4	79.0	78.4	76.8	70.1	62.3	58.8
	2%	WB	51.4	47.3	49.1	53.4	64.0	72.2	76.7	76.7	73.6	67.7	59.7	55.8
		MCDB	52.2	48.3	50.1	54.9	65.2	73.4	78.0	77.7	74.6	68.9	60.2	56.5
	5%	WB	48.7	44.9	47.1	51.4	61.6	70.7	75.4	75.4	72.3	65.1	58.6	53.8
		MCDB	49.8	46.3	48.0	52.8	62.8	72.0	77.0	76.8	73.6	66.4	59.4	55.1
	10%	WB	46.0	42.5	45.2	49.8	59.6	69.2	74.4	74.6	70.9	63.4	57.0	51.7
		MCDB	47.7	44.3	46.3	51.2	60.5	70.6	76.0	76.1	72.4	65.1	58.3	53.4
Mean Daily Temperature Range	5% DB	MDBR	8.0	7.3	6.0	5.1	4.4	4.0	3.7	3.8	4.0	5.4	6.1	7.6
		MCDBR	8.8	8.3	6.9	6.5	5.7	4.6	4.2	4.2	4.1	5.3	5.8	8.5
		MCWBR	10.5	10.4	7.3	5.8	5.2	4.9	5.1	5.0	5.3	8.0	7.4	11.6
	5% WB	MCDBR	8.8	8.4	6.8	6.2	5.0	4.2	4.1	3.5	3.9	4.9	6.0	8.7
		MCWBR	10.8	10.3	8.0	6.3	5.1	5.0	5.2	4.0	4.7	8.1	8.2	11.5
Clear-Sky Solar Irradiance	taub	0.328	0.339	0.362	0.383	0.432	0.472	0.484	0.467	0.400	0.370	0.339	0.329	
	taud	2.474	2.441	2.415	2.385	2.283	2.214	2.186	2.234	2.389	2.470	2.510	2.508	
	Ebn at Noon	264	278	283	283	270	258	253	255	266	263	258	255	
	Edh at Noon	25	30	34	37	42	45	46	43	34	28	24	23	
All-Sky Solar Radiation	RadAvg	586	875	1225	1574	1761	1933	1944	1751	1422	991	680	509	
	RadStd	53	89	106	149	179	184	100	118	116	90	61	46	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (55 neighbors)	+0.72	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+211	+121

Nomenclature: See separate page